THE STACK OF SPACES WITH FP FLAT PROPER FIBRES IS LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition artin-definition-limit-preserving (canonical tag 07XL). cited at statement line 3744. The limit preserving property.
- Lemma spaces-limits-lemma-descend-finite-presentation (canonical tag 07SK). cited at proof line 3751. Colimit of categories of finitely presented spaces.

Source reference. Original definition 07XL: sources/stacks-chow/tags.txt, source lines 10081–10081. Cited at statement line 3744: the definition of limit preserving.

Source reference. Original lemma 07SK: sources/stacks-chow/tags.txt, source lines 9905–9905. Cited at statement line 3751: category of fp spaces is the colimit.

Source reference. Original lemma 08K0: sources/stacks-chow/tags.txt, source lines 10868–10868. Cited at statement line 3758: flatness descends through the limit.

Source reference. Original lemma 08K1: sources/stacks-chow/tags.txt, source lines 10869–10869. Cited at statement line 3759: properness descends through the limit.

Lemma 0.2. *The stack* $p'_{fp,flat,proper} : \mathcal{S}paces'_{fp,flat,proper} \to Sch_{fppf}$ *is limit preserving (Artin's Axioms, Definition 07XL).*

Proof. Let $T = \lim T_i$ be the limits of a directed inverse system of affine schemes. By Limits of Spaces, Lemma 07SK the category of algebraic spaces of finite presentation over T is the colimit of the categories of algebraic spaces of finite presentation over T_i. To finish the proof use that flatness and properness descends through the limit, see Limits of Spaces, Lemmas 08K0 and 08K1. □

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